

ETHICAL AI MODEL DESIGN CHALLENGE

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Course Name: Responsible AI

Case Study: The Cartel That Never Spoke

The case study discusses how companies use AI-based pricing software to automatically decide prices. In markets such as rental housing, multiple businesses may use similar AI systems that analyze market data and recommend optimal prices. Even without direct communication between competitors, these AI systems can gradually set similar high prices, reducing competition and increasing costs for consumers. This creates a challenge for regulators because traditional antitrust laws are designed to detect agreements between humans, not coordination that emerges from interactions between AI systems.

AI Application:

AI-Based Tacit Collusion Detection System

Problem Statement:

Many businesses use AI pricing software to automatically decide prices. When several companies use similar AI systems in the same market, the algorithms may independently increase prices in a coordinated manner without any direct communication. This can reduce competition and lead to higher prices for consumers.

Data Required:

- Market pricing records with timestamps
- Historical price data before AI adoption
- Information about pricing algorithms used by firms
- Data from similar markets without AI pricing tools
- Market factors such as number of competitors, demand, and product similarity

Expected Output:

- A Tacit Collusion Index indicating the possibility of algorithmic price coordination
- Alerts for regulators when suspicious pricing patterns are detected
- Market trend dashboards showing sectors with a high risk of collusion

Ethical Principles:

1. Fairness

The system should protect consumers from unfair price increases and ensure that businesses compete fairly.

2. Privacy

Only necessary market-level data should be collected. Customer information and company trade secrets must remain protected.

3. Transparency

Organizations should clearly disclose when AI systems are being used to influence pricing decisions.

4. Accountability

Companies deploying AI pricing systems must take responsibility for the outcomes produced by those systems, even if the behavior was unintended.

5. Safety & Reliability

The model should accurately identify harmful pricing patterns without generating false accusations against businesses responding to normal market changes.

Ethical Risk:

Accountability Vacuum

A major ethical concern is that consumers may suffer from higher prices, but no single party can be held responsible. Companies may claim they only followed software recommendations, while software providers may argue that they only supplied the technology. This creates a gap in accountability despite real consumer harm.

Mitigation Strategy:

Outcome-Based Algorithmic Pricing Audits

AI pricing systems should be tested before deployment to identify collusive behavior. After deployment, regulators should continuously monitor market outcomes using the Tacit Collusion Index. If anti-competitive pricing patterns are detected, firms should be required to retrain or modify their algorithms under regulatory supervision.

Resources

1. RealPage Litigation (2024) – DOJ and multiple state Attorneys General filed antitrust lawsuits alleging that RealPage’s software enabled landlords to coordinate rent increases.
2. Competition Commission of India, Market Study on AI and Competition (2025) – Discussion on algorithmic collusion and AI-enabled price coordination.
3. NBER Working Paper 34054 (2025) – Dou, Goldstein & Ji, *AI-Powered Trading, Algorithmic Collusion, and Price Efficiency*.
4. German Fuel Station Study (2024) – Research on algorithmic pricing tools and their impact on fuel market pricing.
5. DOJ Statements (2025) – Remarks by Assistant Attorney General Gail Slater regarding algorithmic pricing investigations.